

MECHATRONICS SPECTRUM TEAM

Assembly Test BANK

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ملاحظات هامة:

- يمنع نسخ الملف هذا وتغير شعار Spectrum-Team.

الرمز	الدلالة
*	يتكرر بالسنوات لكن تتغير أرقام السؤال فقط
*	يتكرر بالسنوات لكن تتغير كلمات السؤال فقط
*	يتكرر بالسنوات لكن تتغير خيارات السؤال
*	يتكرر بالسنوات كما هو
*	أسئلة سنوات شاشات
*	أسئلة كويزات

Answered By: Omar Eid

Part. 1 (MID MCQs):

Q1. * Data bus width of Intel 8086 microprocessor is

A) 16
B) 24
C) 20
D) 32

Q2. * The word length of Intel 8086 microprocessor is

A) 24
B) 16
C) 20
D) 32

Q3. * The segment that contains the assembly language instructions to be executed by the microprocessor is

A) Data segment
B) Code segment
C) Stack segment
D) Extra segment

Q4. * If offset is a 10-bit number address, then the maximum possible locations are

A) 1024 KB
B) 1024 B
C) 64 KB
D) 1 MB

Q5. If offset is an 11-bit number address, then the maximum possible location is

A) 8 KB
B) 1 KB
C) 2 KB
D) 4 KB

Q6. * If offset is a 12-bit number address, then the maximum possible location is

A) 2 KB
B) 1 KB
C) 1 MB
D) 4 KB

Q7. * If the size of memory is 64K, what will be the starting and ending offset addresses of it?

A) 0000H to FFFFH
B) 0000H to AFFFH
C) 0000H to FFFFFH
D) 000H to FFFFH

Q8. * If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?

A) 000H to FFH
B) 000H to 1FFH
C) 0000H to FFFFH
D) 000H to FFFH

Q9. * Instruction Pointer (IP) contains offset address of …… segment

A) Data segment
B) Code segment
C) Stack segment
D) Extra segment

Q10. * The offset of a particular segment varies from _____ to _____

A) 000H to FFFH
B) 0000H to FFFFH
C) 00H to FFH
D) 00000H to FFFFFH

Q11. * If the offset address of X in data segment is equal to 105H, and DS = 1000H, then the physical address of X = ?

A) 10104H
B) 10105H
C) 1105H
D) 11015H

Q12. If the offset address of X in data segment is equal to 118H, and DS=2000H, then the physical address of X = ?

A) 21118H
B) 20118H
C) 21117H
D) 31117H

Q13. If the offset address of X in data segment is equal to 111H, and DS=3019H, then the physical address of X = ?

A) 302AH
B) 302A1H
C) 312AH
D) 3039H

Q14. If the offset address of X in data segment is equal to 564H, and DS=1000H, then the physical address of X = ?

A) 15460H
B) 10564H
C) 10546H
D) 100056H

Q15. * One of the following control transfer instruction is not used with signed number

A) JGE
B) JBE
C) JL
D) JLE

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Q16. Which of the following instructions is the same as JB instruction?

A) JNGE
B) JNB
C) JNAE
D) JNA

Q17. * What is the type of addressing mode used in instruction MOV AL,[BP+6]

A) Indirect
B) Based relative
C) Direct Indexed
D) Direct

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Q18. What is the type of addressing mode used in instruction MOV AL,X[SI]

A) Indirect
B) Based relative
C) Direct Indexed
D) Direct

Q19. What is the type of addressing mode used in instruction MOV BL,[BX]

A) Indexed
B) Direct
C) Direct Indexed
D) Indirect

Q20. Which of the following is an example of a based indexed addressing mode?

A) MOV AX,[BX+SI+1CH]
B) MOV AX,[BP+1]
C) MOV AX,[SI+1]
D) MOV AX,[BX]

Q21. * Which is part of execution unit in Intel 8086 microprocessor?

A) Segments registers
B) ALU
C) Physical address generating unit
D) IP register

Q22. * Which of the following flags treats with string operations?

A) DF
B) AF
C) IF
D) TF

Q23. * The equivalent code of NOT AX instruction is

A) ADD AX,1
B) AND AX,FFFFH
C) OR AX,0000H
D) XOR AX,FFFFH

Q24. * If the SP register contains the hexadecimal value F104H, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP BX
```

A) F108H
B) F102H
C) F104H
D) F106H

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Q25. * If the SP register contains the hexadecimal value F11FH, what will be its value after executing the following program segment?

```
PUSH AX
PUSH BX
ADD AX,BX
POP AX
POP AX
PUSH CX
```

A) F11FH
B) F11AH
C) F11CH
D) F11DH

Q26. * The content of register BL after execution the instruction MOV BL,-7

A) BL=F9H
B) BL=FAH
C) BL=FCH
D) BL=FDH

Q27. * The content of register BL after execution the instruction MOV BL,-11

A) BL=F5H
B) BL=F6H
C) BL=05H
D) BL=0BH

Q28. * In the following instruction sequence, show the values of the Carry, and Zero flags

```
MOV AL,00110011b
TEST AL,8
```

A) CF=0 , ZF=0
B) CF=1 , ZF=0
C) CF=0 , ZF=1
D) CF=1 , ZF=1

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Q29. * In the following instruction sequence, show the values of the Carry, and Zero flags

```
MOV AL,00110011b
TEST AL,4
```

A) CF=1 , ZF=0
B) CF=0 , ZF=0
C) CF=0 , ZF=1
D) CF=1 , ZF=1

Q30. * Find the content of AL and ZF register after execution of the following code

```
MOV AL,FCH
TEST AL,16
```

A) AL=FCH , ZF=1
B) AL=FCH , ZF=0
C) AL=00H , ZF=1
D) AL=0CH , ZF=0

Q31. * Find the content of AL and zero flag (ZF) and carry flag (CF) after execution of the following code

```
MOV AL,00H
SHR AL,1
ADC AL,3
```

A) AL=03H , ZF=0 , CF=0
B) AL=00H , ZF=0 , CF=1
C) AL=0CH , ZF=0 , CF=0
D) AL=02 , ZF=1 , CF=1

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Q32. * Find the content of AL and zero flag (ZF) and sign flag (SF) after execution of the following code

```
MOV AL,0FDH
ADD AL,3
ADC AL,0FEH
```

A) AL=FFH , ZF=0 , SF=1
B) AL=02H , ZF=1 , SF=1
C) AL=FFH , ZF=0 , SF=0
D) AL=02H , ZF=0 , SF=1

Q33. * Find the content of AL and zero flag (ZF) after execution of the following code

```
MOV AL,FDH
ADD AL,3
ADC AL,3
```

A) AL=4 , ZF=0
B) AL=2 , ZF=1
C) AL=5 , ZF=0
D) AL=6 , ZF=1

Q34. In the following instruction sequence, show the values of register AL, and Zero flag

```
MOV AL,00110011B
AND AL,F0H
```

A) AL=30H , ZF=0
B) AL=03H , ZF=0
C) AL=30H , ZF=1
D) AL=03H , ZF=1

Q35. * Number of the times the instruction sequence below will loop before coming out of loop is

```
MOV AL,FCH
L: ADD AL,1
JNZ L
```

A) 255
B) 4
C) 2
D) 3

Q36. * Find the value of AX after execution the following assembler codes

```
MOV CX,5
MOV AX,0
L: ADD AX,CX
DEC CX
CMP CX,2
JA L
```

A) 9
B) 15
C) 8
D) 12

Q37. * Find the value of AX after execution the following assembler codes

```
MOV CX,6
MOV AX,0
L: ADD AX,CX
DEC CX
CMP CX,3
JA L
```

A) 9
B) 15
C) 8
D) 12

Q38. * Show the binary value in register AL after execution the following instruction sequence

```
MOV AL,11
MOV BL,15
SUB AL,BL
```

A) AL=11111111
B) AL=11111011
C) AL=11111110
D) AL=11111100

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Q39. Show the binary value in register AL after execution the following instruction sequence

```
MOV AL,7
MOV BL,11
SUB AL,BL
```

A) AL=11111001
B) AL=11110101
C) AL=11111100
D) AL=11111000

Q40. * In the following instruction sequence, show the value of AL, in binary

```
MOV AL,3FH
NEG AL
XOR AL,FFH
```

A) AL=00111110B
B) AL=01011111B
C) AL=01011110B
D) AL=01011111B

Q41. In the following instruction sequence, show the value of AL

```
MOV AL,0FCH
NOT AL
XOR AL,0FFH
```

A) AL=FBH
B) AL=05H
C) AL=FCH
D) AL=06H

Q42. * Find the content of memory location AH after execution of the following code

```
MOV AX,14
MOV BL,4
DIV BL
XOR AH,0FH
```

A) D0H
B) C0H
C) 0DH
D) 0CH

Q43. Find the content of memory location AX after execution of the following code

```
MOV AX,FF02H
MOV BL,4
MUL BL
XOR AX,FFFFH
```

A) FFF7H
B) FFEFH
C) FFF8H
D) 00F7H

Q44. * Suppose X defined as below; what is the content of AX register after execution of the following instructions?

```
X DD 335647FBH,22H
MOV BX,OFFSET X
MOV AL,[BX]
INC BX
MOV AH,[BX+1]
```

A) AX=56FBH
B) AX=FB56H
C) AX=33FBH
D) AX=47FBH

Q45. Suppose X defined as below; what is the content of AX register after execution of the following instructions?

```
X DD 3C0347FBH,20H
MOV BX,OFFSET X
MOV SI,2
MOV AX,[BX+SI]
INC SI
ADD AH,[BX+4]
```

A) AX=2030H
B) AX=0320H
C) AX=3C47H
D) AX=5C03H

Q46. Suppose X defined as below; what is the content of CX and AH registers after execution of the following instructions?

```
X DW 4580H,0ABCH,1100H,2222H
MOV BX,OFFSET X
INC BX
MOV CX,[BX]
INC BX
ADD AH,[BX]
```

A) CX=BC45H , AH=BCH
B) CX=0ABCH , AH=0AH
C) CX=0ABCH , AH=00H
D) CX=BC45H , AH=0CH

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Q47. If AL contains ASCII code of an uppercase letter, it can be converted to lowercase by

A) XOR AL,32
B) AND AL,00100000B
C) OR AL,00100000B
D) SUB AL,32

Q48. Which of the following instruction can be used to multiply constants in register DX by 8?

A) MOV CL,2 ; RCL DX,1
B) MOV CL,2 ; ROL DX,1
C) MOV CL,3 ; SHL DX,CL
D) MOV CL,3 ; SAR DX,1

Q49. The output of the following operation on register BL is

```
MOV BL,FCH
MOV CL,8
ROL BL,CL
```

A) BL=CFH
B) BL=FCH
C) BL=0FH
D) BL=00H

Q50. If 1110 1100B is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

A) -20
B) -11
C) -19
D) -18

Q51. * If the range of physical addresses of a given memory is 0000H–3FFFH, the number of required address bus bits in this memory is

A) 16 bits
B) 14 bits
C) 12 bits
D) 15 bits

Q52. The value of AX at the end of execution of the following 8086 assembly instruction sequence is

```
MOV AX,-9
SAR AX,8
```

A) AX=00F7H
B) AX=0FFFH
C) AX=FF00H
D) AX=FFFFH

Q53. * Assuming DS=4000H, SI=10H, and BX=23H, the physical address of the source operand in the instruction MOV AX,[BX+SI+03H] is

A) 4006H
B) 4FFF6H
C) 40033H
D) 40036H

Q54. * Based in the following code answer the questions

```
Y DW 0,0
MOV BX,200H
MOV SI,1
MOV 4[BX],1133H
MOV AL,[205H]
MOV AH,BH
MOV Y[SI],AX
```

The value of Y is

A) Y DW 1100H,0002H
B) Y DW 11,0200H
C) Y DW 3300H,02H
D) Y DW 0200H,0011H
E) Y DW 1100H,0200H

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Q55. * Which of the following instruction has indirect addressing mode

A) MOV AL,X[BX]
B) MOV AL,[BP]
C) MOV AL,X
D) MOV AL,[SI+2]

Q56. * What is the value of X+2 after executing the following code

```
X DB 12H,10,3
Y DW 33H,2255H
MOV AX,Y+1
MOV X+2,AH
```

A) 3
B) 55H
C) 10
D) 33
E) 0

Q57. * What is the value stored in the location X+6 if define the following data

X DB -4,5
Y DW 22H,-6,344H

A) 06H
B) 44H
C) 22H
D) FFH
E) 00H

Q	1	2	3	4	5	6	7	8	9	10
Answer	A	B	B	B	C	D	A	B	B	B
Q	11	12	13	14	15	16	17	18	19	20
Answer	B	B	B	B	B	C	B	C	D	A
Q	21	22	23	24	25	26	27	28	29	30
Answer	B	A	D	C	D	A	A	C	C	B
Q	31	32	33	34	35	36	37	38	39	40
Answer	A	A	A	A	B	D	B	D	C	A
Q	41	42	43	44	45	46	47	48	49	50
Answer	C	C	A	A	D	A	C	C	B	A
Q	51	52	53	54	55	56	57			
Answer	B	D	D	A	B	B	B			

Part. 2 (MID Code Tracing):

Q1. Define below X memory locations and the offset address of X is 100h. Find the content of register AL and DX after execution of the following code.

```
X DW 35FAH,48CBH,231AH,0080H
MOV BX,OFFSET X
ADD BX,2
MOV AL,[BX]
INC BX
MOV DX,[BX]
```

Q2. Defined below X memory locations and the offset address of X is 100h. Find the content of register AL and BL after execution of the following code.

```
X DB 88H,44H,50H,90H,55H,23H,51H
MOV BX,OFFSET X
MOV SI,0
MOV AL,[BX+4]
MOV CL,X[SI]
```

Q3. After execution of the following code, find AX, CL, CF, ZF, PF

```
MOV CL,03H
MOV AL,127
ROR AL,01H
JNC L
ADC CL,00H
L: SHR AL,CL
```

Q4. After execution of the following code, find AL and AH

```
MOV AL,14
MOV AH,0
MOV BL,17
DIV BL
```

Q5. Find the content of AL register

```
MOV AL,FCH
TEST AL,08H
```

Q6. Trace the following program segment and determine the final value for each register below

```
MOV AX,64FH
MOV BX,0CFH
AND AX,BX
MOV CL,FDH
NEG CL
PUSH AX
MUL CL
PUSH AX
MUL CL
POP DX
POP CX
```

Q7. Trace out the following program segments and find registers contents

```
ARR DW 3333H,4444H,1111H,2222H,0FFFH
MOV BX,OFFSET ARR
MOV SI,4
MOV DL,2
MOV AL,[BX+SI]
MOV AH,[BX]
XCHG AL,DL
```

Q8. Trace out the following program segments and find registers contents

```
ARRAY DB 8,3,2,4,6,8,5,7,0C0H
MOV BX,OFFSET ARRAY
MOV CX,4
L:
ADD AL,[BX]
ADD BX,2
LOOP L
MOV AH,[BX]
```

Q9. Find the outputs of the following program segment

```
X DD 1345700CH
MOV DX,X
MOV BX,3000H
PUSH BX
PUSH DX
TEST DL,BH
POP DX
```

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Q10. Find the outputs of the following program segment

```
X DD 335647FBH,22H
MOV BX,OFFSET X
MOV AL,34H
ADD [BX+1],AL
XOR AL,0FH
```

Q11. Find the final value of AX

```
MOV CX,1
MOV AX,0
L: ADD AX,CX
INC CX
INC CX
CMP CX,8
JB L
```

Q12. Show the binary value in register AL after execution the following instruction sequence

```
MOV AL,8
MOV BL,11
SUB AL,BL
```

Q13. Find the values of AL and CL

```
X DB 88H,44H,50H,90H,55H
Y DW 2000H,7600H,0077H,1933H,5511H,4022H
MOV BP,OFFSET Y
MOV SI,0
MOV AL,[BP-4]
MOV DL,[BP+6]
MOV CL,X[SI]
```

Q14. Find the contents of the registers written below for each of the following assembly codes

```
X1 DW 3000H
X2 DW -27ABH,0CCCCH
X3 DB 11B
MOV AL,X3
PUSH AX
PUSH X1+4
PUSH X1+2
POP DX
POP CX
POP BX
```

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Q15. Find the contents of the registers written below for each of the following assembly codes

```
X DB 1,2,3,4,5,6,7,9
MOV CX,8
MOV SI,0
L: CMP X[SI],6
JA L1
ADD AL,X[SI]
L1: INC SI
LOOP L
```

Q16. Find the final values of BL and DL

```
MOV BL,13H
L: SHR BL,1
INC DL
CMP BL,0
JNZ L
XOR DL,FFH
```

Q17. Find the final values of AX, BX, and CX

```
MOV CX,8
MOV BL,2
MOV AH,0
MOV AL,5
L: MUL BL
CMP AL,0
LOOPNZ L
```

Q18. Find the content of AL register

```
MOV AL,FCH
TEST AL,08H
XOR AL,0FH
```

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Question	Answer
Q1	AL = CBH, DX = 1A48H
Q2	AL = 55H, BL = 00H, CL = 88H
Q3	AL = 0BH, CL = 04H, CF = 1, ZF = 0, PF = 0
Q4	AX = 0E00H
Q5	AL = 0FCH
Q6	AX = 02C7H, BX = 00CFH, CX = 004FH, DX = 00EDH
Q7	AX = 3302H, DX = 0011H
Q8	AX = 0C015H, CX = 0
Q9	BX = 3000H, DX = 700CH
Q10	AL = 3BH
Q11	AX = 0010H, CX = 0009H
Q12	AL = 0FDH
Q13	AL = 44H, CL = 88H, DL = 33H
Q14	AX = 0003H, BX = 0003H, CX = 0CCCCH, DX = D855H
Q15	AL = 15H
Q16	BL = 00H, DL = 0FAH
Q17	AX = 0100H, BX = 0002H, CX = 0000H
Q18	AX = 00F3H

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Part. 3 (MID Code Writing)

Q1. Write assembler codes to push into the stack integer numbers from 1 to 99

Q2. Write assembler codes to push even numbers from 1 to 100 into the stack

Q3. Write assembler codes to test bit#2 in register BL; if it is 1 clear bit#3 and bit#5; else set bit #4.

Q4. Write an assembler code to implement the following if statement
if x < 8 s = s + 15 else s = s + 7 assume x and s are signed bytes.

Q5. Write an assembler code to implement the following if statement
if x < 8 s = s + 15 else s = s + 7 assuming x and y are unsigned bytes.

Q6. Write an 8086 assembler code equivalent to the following C++ code

```
if (a1 <= b1) AND (b2 > a2)
X = 5
else
X = 4
```

Q7. Write an assembler codes to store zeroes into X memory locations

```
X DB 10 DUP (?)
```

Q8. Write instruction sequence equivalent to

```
SUB AX,BX
```

Q9. Write instruction sequence equivalent to

```
NEG AX
```

Q10. Write instruction sequence equivalent to

```
LOOP L
```

Q11. Write instruction sequence equivalent to

```
XCHG AL,BL
```

Q12. Write an 8086 assembly program to find the total number of memory byte locations in the memory array named Array in which the bit 2 is set and store it in the byte memory location named count

```
Array DB 21H,84H,0F7H,0E8H,0ABH,8CH,0C4H,44H,74H,95H
count DB 0
```

Q13. Given `X DB 20,15,30,40,41` write an assembler code to find the sum of each memory locations that have ones in bits 4.

Part. 4 (MID Problems)

Q1. Suppose we have a microprocessor with 32 bits address bus. What is the maximum memory capacity will be?

Q2. Suppose we have a microprocessor with 21 bits address bus. What is the maximum memory capacity will be?

Q3. For the following instruction MOV [BX],50H; what is the addressing mode used?

Q4. For the following instruction MOV [BX], AL; what is the addressing mode used?

Q5. Calculate the physical address of memory location assuming that DS=8C00h and BX=4000h



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Part. 5 (FINAL MCQs)

Q1. * The interrupt vector contains two words (pointers) which specify the address of

A) Interrupt vector table
B) Interrupt service routine
C) Interrupt type
D) I/O ports

Q2. * The physical address of interrupt vector associated with INT 8 is

A) 00016H
B) 00020H
C) 00040H
D) 00015H

Q3. The physical address of interrupt vector associated with INT 16 is

A) 00042H
B) 00040H
C) 00064H
D) 0003CH

Q4. * The interrupt vector for INT 13H is stored in memory at

A) 0012H
B) 0048H
C) 004CH
D) 0084H

Q5. * The size of interrupt vectors table is

A) 1024 bytes
B) 4 bytes
C) 64 bytes
D) 128 bytes

Q6. * The interrupt generated by the execution of an interrupt instruction INT is

A) NMI interrupt
B) Software interrupt
C) External interrupt
D) Interrupt-in-interrupt

Q7. * During the execution of an interrupt the data pushed into the stack is the contents of

A) IP
B) CS
C) Flag register
D) All of them

Q8. * An interrupt breaks the execution of program and diverts it to execution to

A) Interrupt service routine
B) Execution unit
C) Counter word register
D) Control unit

Q9. The last instruction in interrupt service routine should be

A) END
B) ENDS
C) IRET
D) INT

Q10. * During the execution near call instruction the ..... changed

A) Only IP
B) Only CS
C) IP,CS
D) IP,DS

Q11. * At the end of interrupt when the CPU executes IRET

A) Clears the trap flag
B) Contents of IP and CS are retrieved
C) Clears the interrupt flag
D) The control transfers to ISR to main program

Q12. * If offset is a 12-bit number address, then maximum possible locations are

A) 2 KB
B) 1 KB
C) 4 KB
D) 1 MB

Q13. * In 8086 microprocessor the overflow flag is set when

A) The result is more than 16 bit
B) Signed number goes out of their range after an arithmetic operation
C) If CF and SF are set to 1
D) During subtraction

Q14. * If there is a carry from lowest 4 bit during addition, ... flag sets

A) Auxiliary carry
B) Carry
C) Overflow
D) Sign

Q15. * The result of division instruction DIV BL is stored in register pair

A) AH:AL
B) AX:DX
C) CX:AX
D) AX:CX

Q16. The result of multiplication instruction MUL BX is stored in register pair

A) DX:AX
B) AX:DX
C) CX:AX
D) AX:CX

Q17. * The equivalent instructions of STOSB assuming DF flag = 0

A) MOV [DI],AL INC DI
B) MOV [SI],AL INC DI
C) MOV AL,[SI] INC SI
D) MOV [DI],AL DEC DI

Q18. * To compare two strings in equality we use prefix ..... before instruction CMPSB

A) REP
B) REPNE
C) REPE
D) REPNZ

Q19. To compare two strings in unequality we use prefix ..... before instruction CMPSB

A) REP
B) REPNE
C) REPE
D) LOOPNZ

Q20. * Find the content of register AL after execution of the following code

```
MOV AL,0FH
SHR AL,1
ADC AL,1
```

A) 9
B) 8
C) 7
D) 10

Q21. Find the content of register AL after execution of the following code

```
X DB 'ABCD*KLM+A123'
CLD
MOV CX,14
MOV AL,'*'
LEA DI,X
REPNE SCASB
MOV SI,DI
LODSB
```

A) '3'
B) '*'
C) 'D'
D) 'K'

Q22. * Find the content of register AL after execution of the following code

```
MOV AL,4
MOV BL,6
CMP AL,BL
JG L
ADD AL,BL
L: XOR AL,BL
```

A) 02H
B) FBH
C) 0CH
D) 0DH

Q23. Find the content of register AL after execution of the following code

```
MOV AX,FF02H
MOV BL,4
MUL BL
XOR AL,FFH
```

A) 5FH
B) 7FH
C) F7H
D) 8FH

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Q24. * The binary value of AL and CF after the following instructions have executed is

```
MOV AL,10101011B
SAR AL,1
```

A) AL=01010101B , CF=1
B) AL=01010101B , CF=0
C) AL=11010101B , CF=1
D) AL=01010100B , CF=0

Q25. * The binary value of AL and CF after the following instructions have executed is

```
MOV AL,57H
SHL AL,1
```

A) AL=10111110 CF=1
B) AL=10101110 CF=0
C) AL=00101011 CF=1
D) AL=00101011 CF=0

Q26. The binary value of AL and CF after the following instructions have executed is

```
MOV CL,2
STC
MOV AL,5BH
RCR AL,CL
```

A) AL=11010110 CF=1
B) AL=11010110 CF=0
C) AL=00110111 CF=0
D) AL=00110111 CF=1

Q27. The binary value of AL and CF after the following instructions have executed is

```
MOV CL,4
MOV AL,7FH
ROL AL,CL
```

A) AL=11110111 CF=0
B) AL=11110111 CF=1
C) AL=01111111 CF=1
D) AL=01111111 CF=0

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Q28. * The value of AL and CF after the following instructions have executed is

```
MOV AL,44
SHR AL,1
```

A) AL=22 CF=0
B) AL=22 CF=1
C) AL=88 CF=1
D) AL=88 CF=0

Q29. What is the value of AL and CF after the following instructions have executed

```
MOV AL,B6H
STC
RCR AL,1
```

A) AL=DBH CF=1
B) AL=DBH CF=0
C) AL=6DH CF=1
D) AL=6DH CF=0

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Q30. * LOOPNE instruction repeat a group of instructions while

A) CX=0 ZF=0
B) CX≠0 ZF=0
C) CX≠0 ZF=1
D) CX=0 ZF=1

Q31. * The value of CX and AL after executing the following code is

```
MOV AL,FFH
MOV CX,6
L1: ADD AL,1
LOOPZ L1
```

A) CX=4 AL=2
B) CX=2 AL=3
C) CX=5 AL=1
D) CX=**4** AL=**1**

Q32. * The value of CX and AL after execution the following code is

```
MOV AL,FFH
MOV CX,5
L1: ADD AL,1
LOOPNZ L1
```

A) CX=4 AL=0
B) CX=5 AL=0
C) CX=2 AL=1
D) CX=3 AL=1

Q33. * The values of CX and AL after executing the following code is

```
MOV AL,0FFH
MOV CX,4
L1: ADD AL,1
LOOP L1
```

A) CX=3 AL=1
B) CX=1 AL=2
C) CX=**0** AL=**3**
D) CX=2 AL=1

Q34. * In the following instruction sequence, show the value of AL, in binary

```
MOV AL,3FH
XOR AL,FFH
```

A) AL=11000000
B) AL=11010000
C) AL=00001111
D) AL=11110000

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Q35. * In the following instruction sequence, show the value of AL, in binary

```
MOV AL,3FH
NEG AL
XOR AL,FFH
```

A) AL=00111110
B) AL=01011111
C) AL=01011110
D) AL=01011111

Q36. In the following instruction sequence, show the value of AL

```
MOV AL,0FAH
NOT AL
XOR AL,FFH
```

A) FBH
B) 05H
C) FAH
D) 06H

Q37. * After execution of the following code register AL contains

```
MOV AL,'a'
ADD AL,2
AND AL, 11011111B
```

A) 'C'
B) 'a'
C) 'A'
D) 'c'

Q38. * Number of times the instruction sequence below will loop before coming out of loop is

```
MOV AL,FFH
L: ADD AL,1
JNZ L
```

A) 1
B) 4
C) 3
D) 0

Q39. * If 11111011 is the 8-bit 2's complement representation of a number N, what is the decimal value of N

A) -11
B) -5
C) 10
D) 12

Q40. * Find the content of AL, ZF and SF after execution of the following code

```
MOV AL,0FDH
ADD AL,3
ADC AL,0FEH
```

A) AL=FFH ZF=0 SF=1
B) AL=00H ZF=0 SF=1
C) AL=FFH ZF=0 SF=0
D) AL=02H ZF=0 SF=1

Q41. Find the content of AH registers after execution of the following code

```
MOV AX,14
MOV BL,4
DIV BL
XOR AH,0FH
```

A) 00H
B) C0H
C) 0DH
D) 0CH

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Q42. * Find the content of memory location X after execution of the following code

```
X DB ?
MOV AX,5
MOV BL,2
DIV BL
OR AH,30H
MOV X,AH
```

A) X='0'
B) X='2'
C) X='1'
D) X=2

Q43. * Find the content of memory location X after execution of the following code

```
X DB 0FH
MOV AX,11
MOV BL,3
DIV BL
OR AH,30H
AND X,AH
```

A) X=2
B) X=0
C) X='0'
D) X='2'

Q44. * Find the contents of memory location X after execution of the following code

```
X DB 'KLM-ABC'
CLD
MOV AL,'Y'
MOV DI,OFFSET X
MOV CX,3
L: STOSB
LOOP L
```

A) YYYM-ABC
B) KLM-YYY
C) YYY-ABC
D) KLM-ABC

Q45. * What is the value of AX register after executing the following instructions

```
MOV AX,45AFH
TEST AL,1
JNZ L1
ADD AL,1
JMP L2
L1: XCHG AL,AH
L2: HLT
```

A) 01FAH
B) 45FAH
C) 00FBH
D) AF45H

Q46. What will be the final value of AX after execution of the following codes

```
MOV AX,5
MOV CX,4
L1: INC AX
LOOP L1
```

A) 11
B) 10
C) 12
D) 9

Q47. Find the value of AL and AH registers after execution of the following assembler

```
MOV AX,11
MOV BL,3
CALL XXX
MOV CL,AL
MOV DL,AH
HLT
XXX PROC
DIV BL
RET
XXX ENDP
```

A) AL=3 AH=2
B) AL=2 AH=3
C) AL=4 AH=2
D) AL=2 AH=4

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Q48. What is the content of register AL and CL after execution of the following code

```
MOV CL,3
MOV AL,04H
L: ADD AL,CL
INC CL
CMP AL,8
JB L
```

A) AL=7 CL=4
B) AL=11 CL=5
C) AL=11 CL=4
D) AL=16 CL=5

Q49. Assume that X memory location defined as below find the content of DX and BX

```
X DD 13457ABCH
MOV DX,X
MOV BX,235FH
XCHG DL,BH
```

A) DX=7A23H BX=BC5FH
B) DX=1323H BX=7A5FH
C) DX=1323H BX=455FH
D) DX=7A13H BX=A85FH

Q50. Assume that the offset address of X is 100H, find the content of DL

```
X DW 567FH,8AC0H,429CH,0017H
MOV BP,OFFSET X
MOV SI,3
MOV DL,[BP][SI]
```

A) 8AH
B) C0H
C) 42H
D) 00H

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Q51. Find the contents of register AL and DL after execution of the following code

```
X DB 'KLM-ABC'
Y DB 'KLM-YBD'
CLD
MOV SI,OFFSET X
MOV DI,OFFSET Y
MOV CX,7
REPE CMPSB
MOV AL,[SI]
MOV DL,[DI]
```

A) AL='A' DL='Y'
B) AL='-' DL='Y'
C) AL='A' DL='-'
D) AL='B' DL='B'

Q52. * What is the value of SI after executing the following code if DF=0

```
MOV SI,201H
LODSW
```

A) 302H
B) 299H
C) 200H
D) 203H

Q53. * What will be appears on the screen after executing the following code

```
STR DB 'ABCDEF',0DH,'AB'
DB 'XYZ','$'
MOV AH,9
MOV DX,OFFSET STR
INT 21H
```

A) ABXYZF
B) ABXYZ$
C) ABCDEF
D) XYZABF

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Q54. * What will be appears on the screen after executing the following code

```
STR DB 'CDEFGHK',0DH,'AB'
DB 'XYZ','$'
MOV AH,9
MOV DX,OFFSET STR
INT 21H
```

A) CDEFGHAB
B) ABXYZH
C) ABCDEFGH
D) ABXYZHK

Q55. * What will be appears on the screen after executing the following code

```
STR DB 'KLMNC',0DH,'ABmk','$'
MOV AH,9
MOV DX,OFFSET STR
INT 21H
```

A) ABmkC
B) KLMAB
C) ABmkLmk
D) KLMNC

Q56. * What is the value of CX register after execution the following codes

```
STR DB 'ABCDE'
MOV DI,OFFSET STR
MOV CX,5
MOV AL,'C'
REPNE SCASB
```

A) 3
B) 2
C) 1
D) 0

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Q57. What is the value of CX register after execution the following codes

```
STR DB 'AB+DE'
MOV DI,OFFSET STR
MOV CX,5
MOV AL,'+'
REPNE SCASB
```

A) 1
B) 3
C) 4
D) 2

Q58. * The values of array Y after executing the following code is

```
X DB 'ABC+CDK'
Y DB 5 DUP(?)
CLD
MOV SI,OFFSET X
MOV DI,OFFSET Y
MOV CX,3
REP MOVSB
MOV AL,'#'
STOSB
```

A) Y=ABC#
B) Y=ABC+CDK
C) Y=ABC#CDK
D) Y=ABC

Q59. The values of array Y and register AL after executing the following code is

```
X DB 'ABC+CDK'
Y DB 5 DUP(?)
CLD
MOV SI,OFFSET X
MOV DI,OFFSET Y
MOV CX,3
REP MOVSB
LODSB
```

A) Y=ABC AL='+'
B) Y=ABC+CDH AL='C'
C) Y=ABC+CD AL='D'
D) Y=ABC AL='D'

Q60. * What will be stored in Y after executing the following codes

```
X DB '123'
Y DB 'ABCDE'
MOV SI,OFFSET X
MOV DI,OFFSET Y+1
MOV CX,2
REP MOVSB
```

A) 123DE
B) 123
C) A12DE
D) AB12E

Q61. * The value of Y after executing the following code is

```
X DB 2,5,7,4,7,8
Y DB ?
MOV BX,OFFSET X
MOV CX,2
MOV AL,0
L: CALL SUM
LOOP L
MOV Y,AL
HLT
SUM PROC
ADD AL,[BX]
INC BX
RET
SUM ENDP
```

A) 11
B) 08H
C) 0
D) 07H

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Q62. The value of Y after executing the following code is

```
X DB 4,7,8
Y DB ?
MOV BX,OFFSET X
MOV CX,2
MOV AL,0
L: CALL XXX
LOOP L
MOV Y,AL
HLT
XXX PROC
MOV AH,[BX]
INC BX
ADD AL,AH
RET
XXX ENDP
```

A) 11
B) 04H
C) 11H
D) 19

Q63. * What is the value of SI after executing the following code if DF=1

```
MOV SI,300H
LODSB
```

A) 300H
B) 299H
C) 2FFH
D) 301H

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Q64. * Complete the following code to have an equivalent code for repeated if equal for

```
MOV AL,'A'
CLD
L1: INC DI
MOV AL,[DI-1]
LOOPE L1
```

A) SUB
B) TEST
C) SCASB
D) CMP

Q65. * The value that will be stored in AX and BX after executing the following instructions are

```
X DW 10H,20H,30H
MOV AX,X+1
PUSH AX
PUSH X
POP AX
POP BX
```

A) AX=0010H BX=2000H
B) AX=0010H BX=0020H
C) AX=2000H BX=0010H
D) AX=0020H BX=0010H

Q66. * The equivalent instructions of MOVSB assuming DF=0

A)	MOV AL,[SI]	MOV [DI],AL	INC DI	INC SI
B)	MOV AL,[DI]	MOV [SI],AL	INC DI	INC SI
C)	MOV AL,[SI]	MOV [DI],AL	DEC DI	DEC SI
D)	MOV AL,[SI]	MOV [DI],AL	INC DI	DEC SI

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Q67. * The register that can be used as a default destination register storing the result of 8-bit multiplication instruction is

A) BX
B) CX
C) AX
D) DX

Q68. Determine the value of register AL after execution of the following assembler codes

```
X DB '123456'
CLD
LEA SI,X
LODSB
MOV BL,AL
LODSB
XOR AL,BL
```

A) AL=03H
B) AL=32H
C) AL=02H
D) AL=33H

Q69. To read 8-bit data from input port with address 3FAH the following instructions can be used

A) MOV DX,3FAH
B) MOV DX,3AFH
C) MOV BX,3AFH
D) MOV BX,3FAH

Q70. * To transfer value 45H 8-bit data from input port addressed by 0CF5H to register AL in microprocessor, the following instructions can be used

A) MOV DX,0CF5H IN AL,DX
B) MOV BX,0CF5H IN AL,BX
C) MOV DX,0CF5H MOV AL,45H OUT DX,AL
D) MOV BX,03FCH OUT DX,AL

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Q71. To transfer value 6F17H 16-bit data from microprocessor to output port with address 3F0H the following instructions can be used

A) MOV DX,03F0H MOV AX,6F17H OUT DX,AX
B) MOV DX,03F0H MOV BX,6F17H OUT DX,BX
C) MOV DX,03F0H MOV AX,6F17H OUT DX,AL
D) MOV DX,03FCH OUT DX,AL

Q72. * All the functions of the ports of 8255 are achieved by programming the bits of an internal register called

A) Address register
B) Read logic control
C) Control word register
D) Data bus control

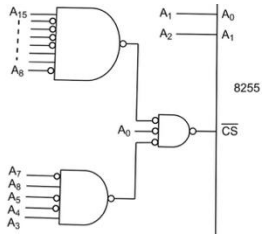
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Q73. * How many bits of data can be transferred between the 8255 PPI and the interfaced device at time

A) 16 bits
B) 24 bits
C) 8 bits
D) 20 bits

Q74. Use the following circuit of interfacing an 8255 with 8086, what is the address of port B

A) 8648H
B) 8649H
C) 7982H
D) 7981H



Q75. In asynchronous serial data transmission, assume the data format is 8 data bit, 1 start bit, 1 stop bit and even parity. What is the speed of transmission baud rate if the device transmit 20 characters per second

A) 220
B) 20
C) 180
D) 200

Q	1	2	3	4	5	6	7	8	9	10
Answer	B	B	B	C	A	B	D	A	C	A
Q	11	12	13	14	15	16	17	18	19	20
Answer	B	C	B	A	A	A	A	C	B	A
Q	21	22	23	24	25	26	27	28	29	30
Answer	D	C	C	C	B	A	B	A	B	B
Q	31	32	33	34	35	36	37	38	39	40
Answer	D	A	C	A	A	C	A	D	B	A
Q	41	42	43	44	45	46	47	48	49	50
Answer	C	C	A	C	D	D	A	B	A	A
Q	51	52	53	54	55	56	57	58	59	60
Answer	D	D	A	D	A	B	D	A	A	C
Q	61	62	63	64	65	66	67	68	69	70
Answer	D	A	B	C	A	A	C	A	A	A
Q	71	72	73	74	75					
Answer	C	C	B	A	A					

Part. 6 (FINAL Code Tracing):

Q1. Find the outputs of the following program segments

```
X DD 1345700CH
MOV DX,X
MOV BX,200FH
PUSH BX
PUSH DX
TEST DL,8H
POP DX
```

Q2. Find the outputs of the following program segments

```
X DW 'KLM+ABC'
Y DW 'KLM+WBD'
CLD
MOV SI,OFFSET X
MOV DI,OFFSET Y
MOV CX,100
REPE CMPSW
MOV AX,[SI]
MOV DX,[DI]
PUSH AX
PUSH DX
POP BX
```

Q3. Find the outputs of the following program segments

```
MOV CX,10
MOV AX,0
L: ADD AX,CX
CMP AX,33
LOOPNZ L
```

Q4. Find the outputs of the following program segments

```
MOV AL,FEH
MOV BL,0
MOV CL,4
ADD AL,2
RCR AL,CL
ADC BL,3
```

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Q5. Find the outputs of the following program segments

```
CLD
STR DB "ABCDE*+"
MOV DI, OFFSET STR
MOV SI, DI
MOV CX, 7
MOV AL, 'D'
REPNE SCASB
MOV SI, DI
LODSB
STOSB
```

Q6. Find the outputs of the following program segments

```
MOV CL, 4
MOV AX, 20FEH
TEST AX, 16
JZ L1
ROR AL, CL ; 00000010 11101111
SHR AH, CL ;
JMP L2
L1: SHL AL, CL
ROL AH, CL
L2: NOT CL
```

Q7. Find the outputs of the following program segments

```
ARRAY DB 4,2,7,9,4,5,5
STC
MOV AL, 0
MOV CX, 4
MOV AH, 0
MOV SI, OFFSET ARRAY
L: ADC AL, [SI]
ADD SI, 1
LOOP L
PUSH AX
PUSH CX
POP CX
```

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Q8. Find the outputs of the following program segments

```
X DD 13457ABCH
MOV DX, X+1
MOV BX, 00FFH
PUSH BX
PUSH DX
XCHG DH, BH
POP DX
POP AX
```

Question	Answer
Q1	BX = 200FH, DX = 700CH
Q2	AX = 0043H, BX = 0044H, CX = 97, DX = 0044H
Q3	AX = 55, CX = 0
Q4	AX = 10H, BL = 3, CL = 4, CF = 0
Q5	AL = 'E', CX = 3, STR = 'ABCDE*+', SI=DI=STR+5
Q6	AX = 02EFH, CL = 0FBH
Q7	AX = 0017H, CX = 0
Q8	AX = 00FFH, BX = 45FFH, DX = 457AH

Part. 7 (FINAL Code Writing)

Q1. Write instructions to test bit-4 in register AL, if it is one reset bit-1 otherwise set bit-2

Q2. Write instructions to test bit-5 in register AL, if it is one reset bit-3 else set bit-2

Q3. Write a procedure to store 1 into 1 KB memory locations defined as byte
X DB 1024 DUP(?)

Q4. Write a procedure to store 0 into 1 KB memory locations defined as byte
X DB 1024 DUP(?)

Q5. Write a procedure to find number of ones in register AL and return the result value into register DL

Q6. Implement the following C++ statement using Intel 8086 assembler codes
If (x>8) s=s+x else s=s-x
s, x memory locations defined as signed numbers

Q7. Write the equivalent 8086 assembly code for
if x > y s=x+y else y++
Assume X, Y, S signed byte memory locations

Q8. Write an 8086 assembler code that is equivalent to the following pseudo C++ code all variables are unsigned 16-bit integers
if a1 <= b1 and b2 > a2
X = 5
else
X = 4

Q9. Using BIOS interrupt, and DOS interrupts write assembler codes to print on the screen
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in position (15,40)

Q10. Using BIOS interrupt write assembler codes to clear screen and set cursor to position (10,20)

Q11. Write an assembly code to move cursor to position (12,30) using BIOS interrupt

Q12. Write an assembler code to reverse string X in order using LODSB and STOSB strings where
X DB 'ABCDEK+++'

Q13. Write instructions sequence equivalent
LOOP instruction

Q14. Write instructions sequence equivalent
NEG AX

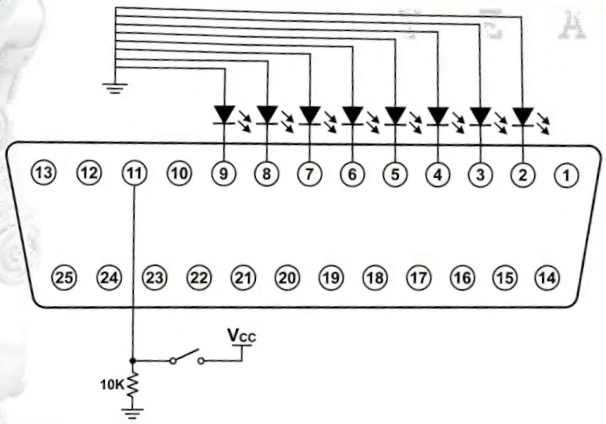
Q15. Write instructions sequence equivalent
MOVSB

Q16. Interface 8-bit input port 74LS245 to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X. Assume the address of the port is 08F0H.

A) Draw block circuit diagram circuit, showing address decoding logic, and control signals
B) Write a program to read the status of switches and store the result into X memory location

Q17. Interface 8-bit input port 74LS245 to read the status of switches SW1 to SW8 to the microprocessor system. Store the status in memory location X defined as byte. Assume the address of the port is 08FFH.

Q18. Based on the figure illustrating eight LEDs connected to the parallel port pins P2–P9 associated with the data register 378H and one switch connected to the parallel port pin P11 associated with the seventh bit of the status register 379H, write an assembly program that turns on the lower four LEDs when the switch is pressed and turns on the upper four LEDs when the switch is not pressed



Part. 8 (FINAL Problems)

Q1. List 3 differences between Synchronous and Asynchronous serial communication
Q2. Explain 2 differences between Asynchronous and Synchronous serial communication
Q3. If the device can transmit 10 char per second find the baud rate, if Data format: 1 start bit, 7 data, 1 parity, 2 stop bits
Q4. Interface 8-bit input port 74LS245 to read the status of switches SW1 to SW8. The switches, when shorted, input a 1 else input a 0 to the microprocessor system. Store the status into register DL. Assuming the address of the port is 0840H.

A) Draw schematic diagram
B) Write assembly code to read status of switches and store the result into DL register



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